

Conditional and Loops

Conditionals

Allow a program to make decisions and execute specific blocks of code based on whether a condition is true or false

For example, assigning grades (A, B, C) based on marks obtained by student

1. if the percentage is above 90, assign grade A
2. if the percentage is above 75, assign grade B
3. if the percentage is above 65, assign grade C

Core Conditional Statements

1. The if Statement

The simplest form of decision-making, it executes a block of code only if the condition is true. [[1](#), [2](#)]

Syntax:

```
if (condition) {  
  
    // code to execute if condition is true  
  
}
```

Use code with caution.

2. The if-else Statement

This allows you to execute one block of code if a condition is true and a different block if it is false. [[1](#), [2](#)]

Syntax:

```
if (condition) {  
  
    // code if true  
  
} else {  
  
    // code if false  
  
}
```

3. The if-else if-else Ladder

Used to test multiple conditions in sequence. Once a condition is met, its block is executed, and the rest of the ladder is skipped. [[1](#), [2](#), [3](#), [4](#)]

Syntax:

```
if (condition1) {  
  
    // block 1  
  
} else if (condition2) {  
  
    // block 2  
  
} else {  
  
    // block if no conditions match  
  
}
```

Cin

In C++, **cin** is an object of istream class that is used to accept the input from the standard input stream i.e. stdin which is by default associated with keyboard. The extraction operator (>>) is used along with cin to extract the data from the object and insert it to the given variable.

Let's take a look at an example:

```
#include <iostream>  
using namespace std;  
int main() {  
  
    // Variable to store data given by cin  
    int a;  
    // Take input using cin  
    cin >> a;  
    cout << a;  
    return 0;  
}
```

While Loop

Definition

A **while** loop is used to **repeat a block of code** as long as a given condition is **true**.

Syntax

```
while(condition) {  
    // code to be executed  
}
```

How it Works

1. First, the **condition is checked**
2. If **true** → **loop runs**
3. After execution → condition is checked again
4. This repeats until condition becomes **false**